

Electrical Engineering Level 6 Wiring Diagrams

1. Ring and Radial Circuit Wiring Diagrams

- Ring Circuit: Used in residential wiring for power outlets; it forms a loop back to the consumer unit.

- Radial Circuit: Each socket or outlet is connected back to the consumer unit without looping.

Diagram: Shows the wiring paths with live, neutral, and earth connections.

2. Multi-way Lighting Circuits

- Two-Way Switching: Allows control of a light from two locations.

- Intermediate Switching: Adds a third (or more) switch to the two-way circuit.

- Three-Way Switching: Similar to intermediate but specific to larger lighting setups.

Diagram: Highlights the connections between switches and lights.

3. Distribution Board (Consumer Unit)

Layout showing circuit breakers (MCB), Residual Current Devices (RCD), and Main Switches.

Diagram: Labels each section for easy identification of wiring routes and protection points.

4. Socket Outlet Wiring

- Standard Socket: A typical 3-pin outlet with connections for live, neutral, and earth.
- Switched Socket: Same as standard, but with an integrated switch to control power.

Diagram: Includes a breakdown of how live, neutral, and earth wires are connected within each socket.

5. Industrial Control Systems Wiring

- Motor Starters: Diagrams for DOL (Direct-On-Line), Star-Delta starters, or Soft Starters.
- Contactors and Relays: Often used for controlling large electrical loads.
- PLC Connections: Used for automation, showing input and output connections.

Diagram: Shows wiring layouts for control circuits and load circuits.

